
Customer Shopping Behavior Analysis

An in-depth transactional behavior report of 3,900 purchases to drive data-centric product positioning, customer loyalty, and targeted marketing strategy.

01 / Exploratory Foundations

Dataset properties, pipeline cleanups, and initial exploratory findings of the consumer demographics database.

Exploratory Data Analysis

Rigorous data preparation pipeline completed in Python to establish structural and relational integrity of 18 transactional dimensions:

-  **Dataset Integrity:** Cleansed transactional registry consisting of 3,900 rows with normalized feature structures.
-  **Missing Data Resolution:** Imputed 37 missing ratings based on median indices calculated dynamically across categories.
-  **Standardization:** Feature columns standardized to uniform snake_case models for optimized query mapping.
-  **Redundancy Control:** Validated overlap of promotional usage metrics and removed highly redundant dimensions.

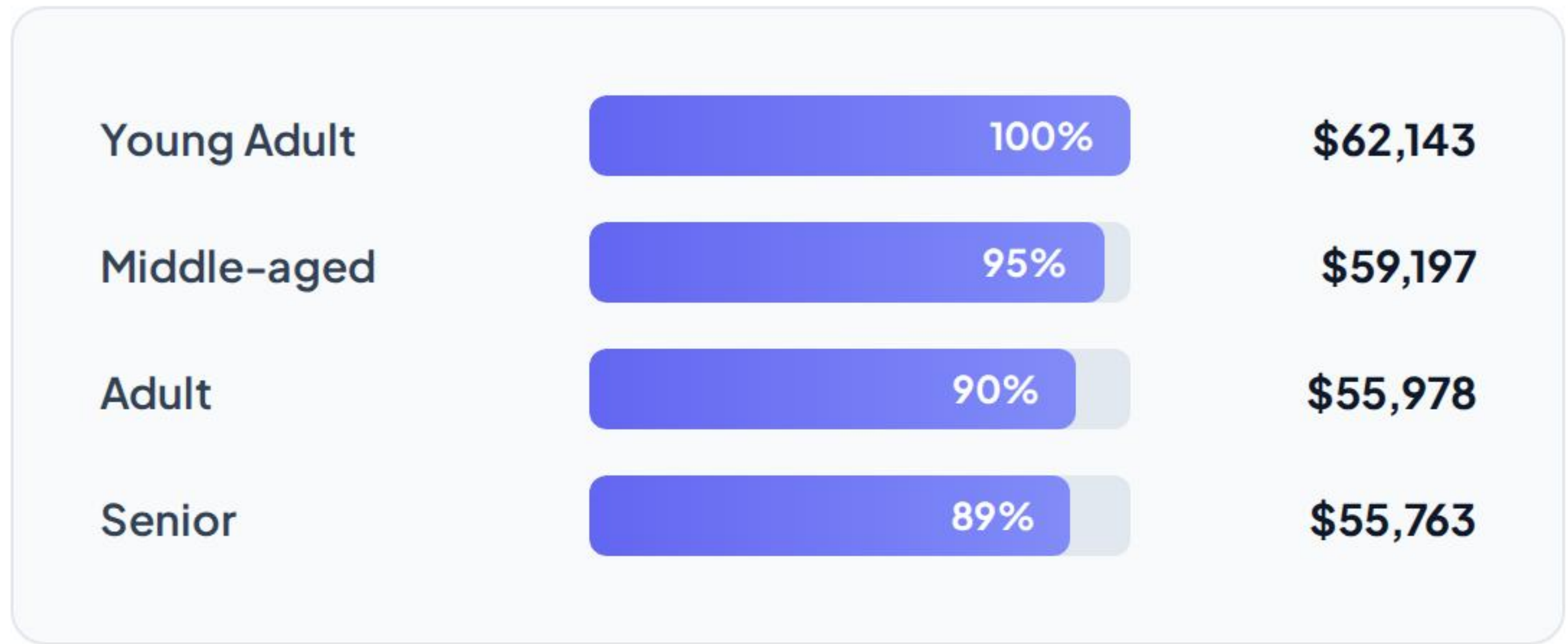
DEMOGRAPHICS FORMULA HIGHLIGHT

We formulate the standard representation of Mean Transaction Purchase Amount ($\$ \overline{x} \$$) using standard MathML markup notation:

$$\bar{X} = \frac{\sum_{i=1}^N x_i}{N}$$

Revenue by Age & Demographic Split

Total Revenue Contribution by Age Group



Demographic Highlights

-  **Gender Spend Disparity:** Males significantly outspend females, generating \$157,890 in revenue compared to \$75,191.
-  **Age Consistency:** Purchasing power remains exceptionally distributed across all binned cohorts with "Young Adults" capturing the lead.

Subscription Status Dynamics

SUBSCRIPTION STATUS	TOTAL CUSTOMERS	AVERAGE SPEND (USD)	TOTAL CUMULATIVE REVENUE
Yes (Subscribed)	1,053	\$59.49	\$62,645.00
No (Unsubscribed)	2,847	\$59.87	\$170,436.00

Key Insight: Although subscription status guarantees customer lifetime values, unsubscribed buyers currently constitute 73% of transactional volume, indicating significant immediate monetization potential.

Discount Dependencies & Delivery Dynamics

Shipping Type Behavior

DELIVERY TIER	AVERAGE SPEND VALUE
Express Delivery	\$60.48
Standard Delivery	\$58.46

Express tier buyers exhibit higher transactional densities, yielding an average spend increase of 3.45% over standard shipping models.

Top 5 Discount-Dependent Products



| Top 3 Products by Category



Accessories

Leading segments with steady high-margin volume metrics:

1. **Jewelry** (171 Orders)
2. **Sunglasses** (161 Orders)
3. **Belt** (161 Orders)



Clothing

The highest cumulative volumetric output segment:

1. **Blouse** (171 Orders)
2. **Pants** (171 Orders)
3. **Shirt** (169 Orders)



Footwear & Outerwear

Consistently high ticket categories with regular purchases:

1. **Jacket** (Outerwear, 163 Orders)
2. **Sandals** (Footwear, 160 Orders)
3. **Coat** (Outerwear, 161 Orders)

Customer Segmentation Portfolio

CUSTOMER SEGMENT CLASSIFICATION	TOTAL SEGMENT COUNT	PERCENTAGE CONTRIBUTION (%)
Loyal Customers	3,116	79.90%
Returning Customers	701	17.97%
New Customers	83	2.13%

Analysis: A massive 79.9% base classified as loyal suggests strong retention indicators. The growth effort must focus on accelerating the acquisition of the "New Customers" pipeline.

Repeat Buyers & Subscriptions

Top 5 Rated Products



High Frequency Repeat Cohorts

Analysis on subscribers with >5 cumulative lifetime purchases reveals an underlying retention risk:

2,518

UNSUBSCRIBED

958

SUBSCRIBED

72% of repeat high-value buyers are currently unsubscribed, representing a key upgrade opportunity.

| Power BI Behavioral Dashboard

Interactive visual insights captured via our newly created Power BI BI interface:

-  **Key Customer Metrics:** Tracks 970 high-volume localized customers.
-  **Spend Indicators:** Average transaction value stable at \$1.248K.
-  **Satisfaction Ratings:** Exceptional average product score of 4.2 stars.
-  **Category Revenue:** Top overall performance dominated by the Clothing and Accessories segments.



600 × 400

Strategic Growth Roadmaps



1. Monetize Subscriptions

Incentivize the 2,518 unsubscribed repeat buyers with tailored exclusive discounts, membership tiers, and customized member benefits.



2. Pricing Optimization

Reduce promotion exposure on discount-dependent baseline items (e.g., Hats, Sneakers) to systematically reclaim standard retail profit margins.



3. Demographic Targets

Focus marketing campaigns towards high-value male cohorts and young adults, emphasizing premier-rated items (e.g., Gloves, Sandals).

Questions & Answers

Review analytical methodology, database cleanups, and key implementation guidelines.

 analytics@corporatedeck.com  www.analyticsportal.com